# Course Roadmap

## MSIS-2402 — Database Management Systems: Fundamentals of SQL

- Santa Clara University · Leavey School of Business
- 2 units · 5 weeks · 10 sessions of 95 minutes · Winter 2027
- Instructor: Dr. Mahmoud Parsian

# 🌍 Course Vision

Transform students from:

❌ spreadsheet users

➡️

✅ data thinkers who can query and explore data

# 🎯 Learning Outcomes

Students will be able to:

- Explain how relational data is organized (tables, keys, relationships)
- Create, retrieve, filter, summarize, and modify data with SQL
- Combine related tables and break complex questions into smaller queries
- Use SQL for data analysis, including window functions and CTEs
- Describe how databases fit into web and other modern applications
- Recognize the main types of modern data systems and the operational, security, performance, and governance issues in managing data

# 🧠 Learning Progression

```
Data

→ Query & Summarize          (Week 1)

→ Combine & Decompose        (Week 2)

→ Analyze                    (Week 3)

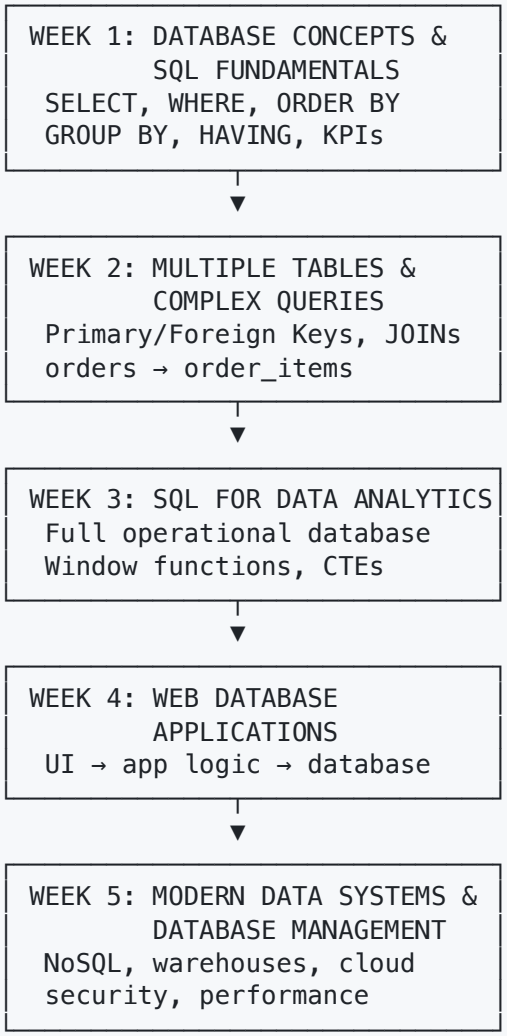
→ Build Applications         (Week 4)

→ Manage Data at Scale       (Week 5)
```

# 📊 Course Structure — 5 Weeks

| Week | Theme | Skill |
|---|---|---|
| 1 | Database Concepts and SQL Fundamentals | Querying & summarizing |
| 2 | Working with Multiple Tables and Complex Queries | Combining & decomposing |
| 3 | SQL for Data Analytics | Insight discovery |
| 4 | Introduction to Web Database Applications | Application architecture |
| 5 | Modern Data Systems and Database Management | The data landscape |

# MSIS-2402 — Course Roadmap



# Slide — Overall Course Progression

- Week 1 → Database Concepts and SQL Fundamentals
- Week 2 → Working with Multiple Tables and Complex Queries
- Week 3 → SQL for Data Analytics
- Week 4 → Introduction to Web Database Applications
- Week 5 → Modern Data Systems and Database Management

## Goal of the course:

Learn how to **analyze real business data using SQL**, and understand how databases power applications and organizations.

# 🟦 Week 1 — Database Concepts and SQL Fundamentals

How relational data is organized, and the core skills for creating, retrieving, filtering, summarizing, and modifying data.

Topics

- What is a database / a relational database / an RDBMS
- What is SQL
- Tables, rows, columns

## 🟦 Week 1 — Retrieving & Filtering Data

Students learn:

- SELECT … FROM
- WHERE, AND / OR / NOT
- ORDER BY
- LIMIT
- DISTINCT

Dataset:

Single **sales** table.

# SQL for Business Analytics

Data

→

1. Transform Data

→

2. Insight

→

3. Decisions

# Week 1 — Example Query

Largest sales in the dataset.

```sql
SELECT *
FROM sales
ORDER BY total_amount DESC
LIMIT 10;
```

Learning outcome:

Students learn to **retrieve and filter data**.

## SQL Mindset

> SQL is NOT code

> SQL is a **question language**

# 🟦 Week 1 — Summarizing Data

- `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`
- `GROUP BY`, `HAVING`
- basic business metrics (KPIs)

```sql
SELECT product_category,
       SUM(total_amount) AS total_sales
FROM sales
GROUP BY product_category
ORDER BY total_sales DESC;
```

## Business Insight

- Which product category leads?
- Where to invest?
- What drives growth?

# Week 2 — Multiple Tables and Complex Queries

Students move from **one table → multiple tables**.

**Tables introduced:**

- `customers`
- `orders`
- `products`
- `order_items`

**Topics**

- primary keys and foreign keys
- relational modeling and normalization
- INNER JOIN and LEFT JOIN
- multi-table joins (and why a wrong `ON` condition silently mismatches rows)

# Week 2 — Example Query

Orders by country.

```
SELECT c.country,
       COUNT(o.order_id) AS num_of_orders
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.country;
```

Learning outcome:

Students learn to **combine tables to answer questions**.

# Week 2 — Example Query

Revenue by product category.

```
SELECT p.category,
       SUM(oi.sales_amount) AS total_revenue
FROM order_items oi
JOIN products p
  ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY total_revenue DESC;
```

Learning outcome:

Students perform their **first multi-table business analytics queries**.

# Week 3 — SQL for Data Analytics

Students explore a **real business schema**.

Tables

```
customers
orders
order_items
products
stores
payments
shipments
```

Topics

- operational analytics and data quality checks
- Window functions and ranking queries
- Common Table Expressions (CTE)

# Week 3 — Example Query

Revenue by store type.

```
SELECT s.store_type,
       SUM(oi.sales_amount) AS total_revenue
FROM stores s
JOIN orders o ON s.store_id = o.store_id
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY s.store_type;
```

Learning outcome:

Students analyze a **complete business database**.

# Week 3 — Example Query

Top customers by revenue.

```sql
SELECT customer_id,
       SUM(sales_amount) AS revenue,
       RANK() OVER (
          ORDER BY SUM(sales_amount) DESC
       ) AS rnk
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY customer_id;
```

Learning outcome:

Students perform **real analytics workflows in SQL**.

# Week 4 — Introduction to Web Database Applications

The role of databases in web and other modern applications.

Topics

- Application architecture: user interface → application logic → database
- Client–server databases
- Basic web technologies
- A simple database-backed application: how data moves through a real application

# Week 5 — Modern Data Systems and Database Management

Beyond SQL and relational databases.

Topics

- Types of data systems and database technologies
- Data platforms (warehouses, lakes, cloud)
- Operations: transactions and concurrency
- Security, performance, and governance of organizational data

## What You Will Be Able To Do

By the end of the course students can:

- explore unfamiliar databases
- write complex SQL queries
- join multiple tables
- perform data analytics using SQL
- answer real business questions
- explain how applications and organizations use and manage databases

SQL becomes a **data analysis tool**, not just a query language.

# 🧪 Tools & Technologies

- SQL
- Python
- **DuckDB** — the main SQL engine (Weeks 1–3)
- **MySQL** — a client–server database (Weeks 4–5)
- **Marimo** — reactive Python notebooks with SQL cells
- **qStudio** — a desktop SQL editor
- CSV datasets

# 🗓️ How Each Session Works

| Time | What happens |
|---|---|
| First ~55 minutes | Lecture and live demo |
| Last ~40 minutes | **Lab A** in class — submit before class ends |
| By **11:59 pm Pacific Time the next day** | Finish and submit **Lab B** on Camino |

Every session has a **Lab A** (core, 40 points, 6 tasks, done in class) and a **Lab B** (a step more challenging, 30 points, 6 tasks) — published in the course repository's `LABS/` folder during class.

# 📦 Assessments

| Component | Points |
|---|---|
| 🧪 10 Labs — one per session, 70 points each (a 40 + b 30) | 700 |
| ✏️ Final Exam — 2 hours, closed book, lock-down browser, after Session 10 | 300 |
| **Total** | **1,000** |

Details: `course_information/ASSIGNMENTS_and_GRADING.md`

# 💼 Teaching Philosophy

- Learn by doing
- Focus on intuition over theory
- Emphasize business questions

# 🎯 Final Outcome

Students will:

- confidently query data
- understand datasets
- be ready for **MSIS-2407** — DBMS - Design, Development & Administration

# 🚀 Career Relevance

Prepares students for:

- Data Analyst
- Business Analyst
- Analytics roles

# 🙌 End of Roadmap

# Let’s Begin

Week 1 starts with a simple question:

**What can we learn from sales data?**

Let’s start querying.